

CURRICULUM OPTIMIZATION FOR HIGH-YIELD TECH STACKS

Stack Overflow Developer Survey Analysis, 2024 - 2025

Sara AlNajjar

General Assembly Data Science - DSBFT2

Introduction & Background

1

Every coding bootcamp risks time and resources to guess which tech skills will land you a job

Limited weeks mean not every language can be taught, choices have real tradeoffs.

2

The largest annual developer survey

Stack Overflow surveys working developers worldwide on what they use and what they earn.

3

Two full survey years analyzed

2024 and 2025 responses give a real, current picture of the market.

Problem Statement & Project Goals

“Which programming languages are associated with the highest pay and the strongest growth? (the ones most worth teaching)”

GOAL: Give the bootcamp a data-backed shortlist of languages to prioritize in its curriculum.

NOTE: Did not want to recommend a language based only on a high salary figure

Data & Methodology



Source

Stack Overflow Developer Survey, 2024 and 2025



Scope

Narrowed to United States respondents to keep pay comparisons fair



Columns

Response ID, Country, Languages Worked With, Converted Compensation Yearly, Education Level, Survey Year



Focus

Language usage matched against reported annual compensation

Data Cleaning and Preparation



Removed responses that did not contain a programming language or an annual compensation value



Excluded outlier compensation



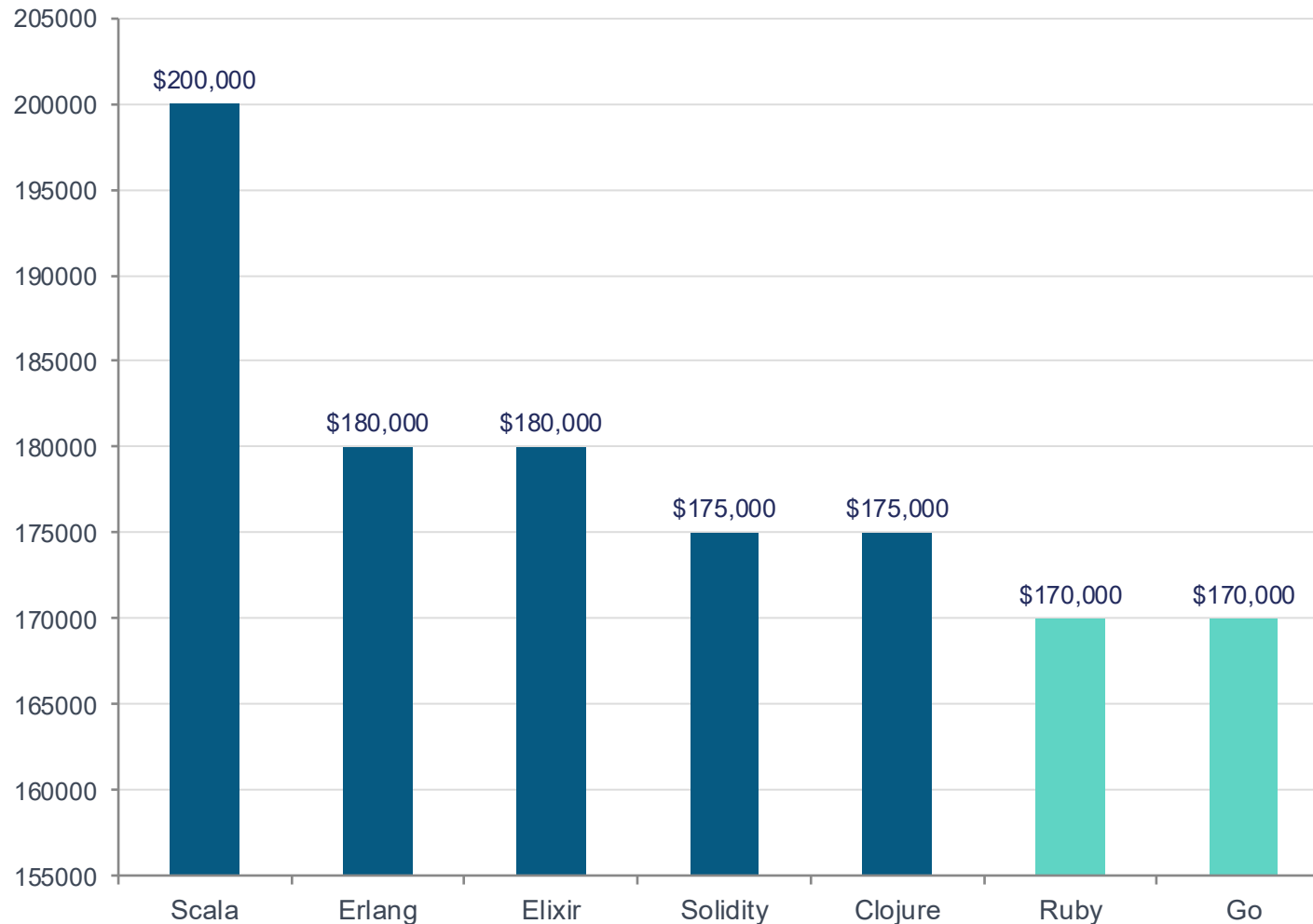
Separated that cell so that each language could be analysed individually

Agenda

- 1 Which languages pay the most
- 2 Which languages are growing the fastest
- 3 Which languages are both high-paying and widely used
- 4 What this means for curriculum

Deep Dive: Highest-Paying Languages

Median annual compensation, US respondents (min. 30 respondents)

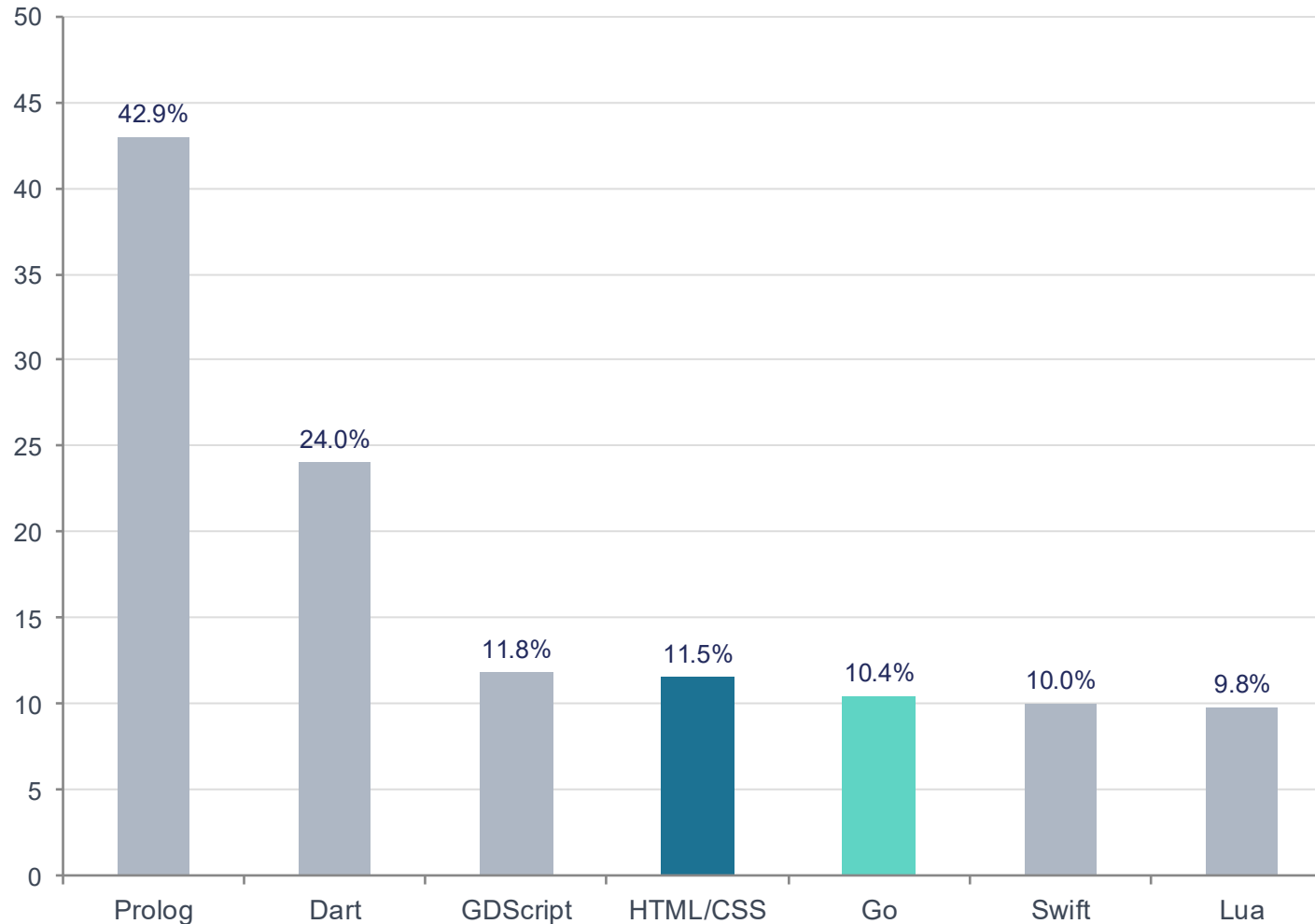


Scala, Erlang, Elixir, Solidity, Clojure: small, specialized respondent groups, not a broad market.

Ruby & Go: the only two backed by hundreds of respondents (890 and 1,692).

Deep Dive: Fastest-Growing Languages

Year-over-year change in median compensation, 2024 - 2025



Reading the growth

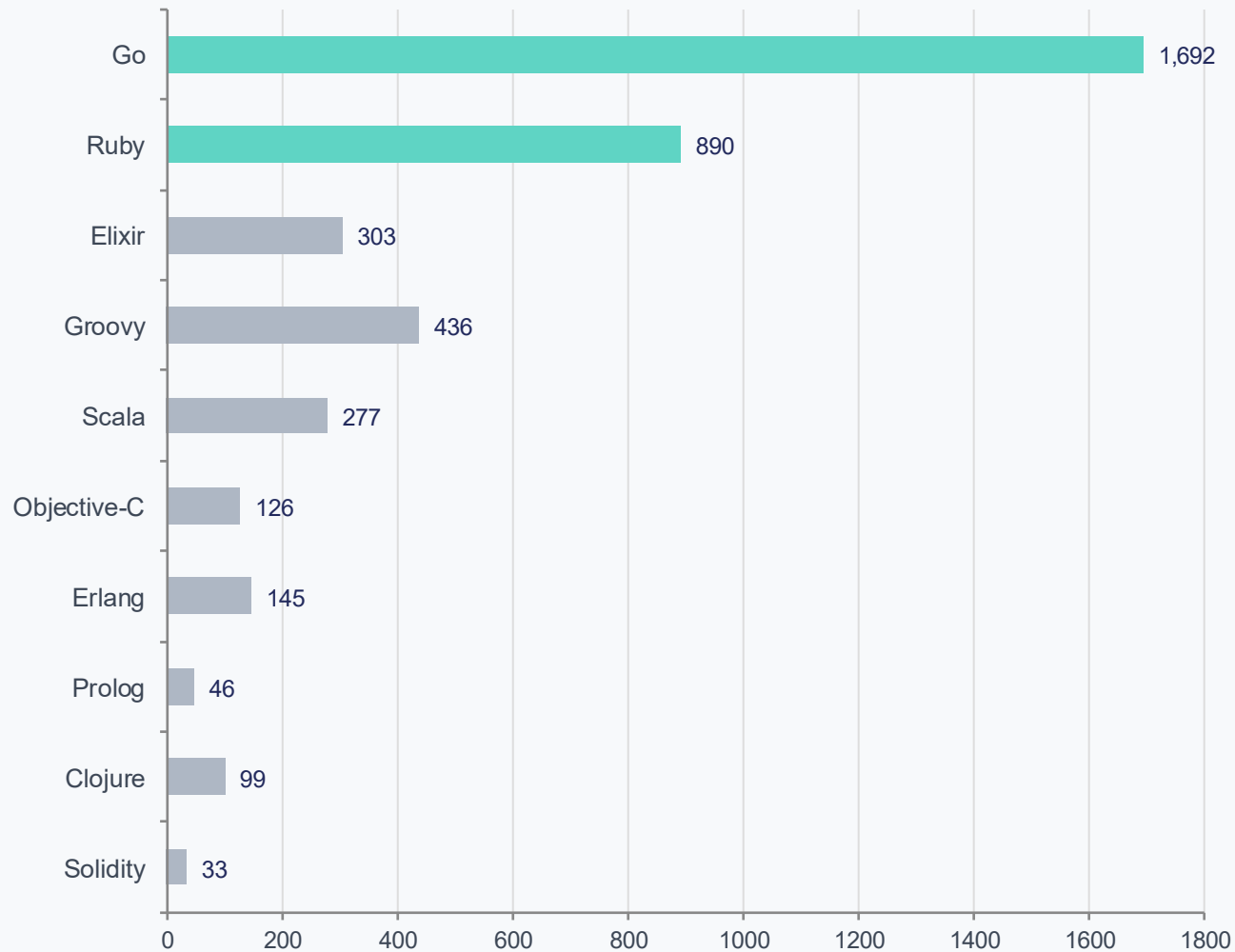
Prolog & Dart: biggest jumps, but small samples make them less dependable.

Go: smaller but far more dependable growth, backed by a large group.

HTML/CSS: a beginner staple, still gaining value.

Analysis & Key Insights

Number of respondents by language, among the top-paying languages (min. 30 respondents)



HIGHEST PAY

Scala: \$200K

Erlang, Elixir, Solidity & Clojure also rank high, but all on small, specialized respondent pools.

FASTEST GROWTH

Prolog +42.9% · Go +10.4%

Prolog's jump is a small-sample outlier; Go's smaller growth is far more dependable, backed by a large group.

THE CORE FINDING

Go & Ruby

The only languages combining high pay with genuinely widespread use. Go stands alone (the only one also growing).

Summary of Findings

#1

Go

Pay + adoption + growth

#2

Ruby

Pay + adoption

\$200K

Scala (top pay)

Narrow market only

Highest pay ≠ best curriculum choice

if adoption is too narrow.

Niche languages (Scala, Erlang, Elixir, Solidity, Clojure) pay well but reach a narrow market.

HTML/CSS remains valuable as a growing, foundational skill.

Recommendations & Next Steps

RECOMMEND

- Make Go a core curriculum priority
- Keep Ruby as a strong secondary offering
- Keep HTML/CSS as a foundational early module
- Offer Scala / Erlang / Elixir / Solidity / Clojure as an advanced elective track

NEXT STEPS

Extend this analysis with job posting volume and experience-level breakdowns to further validate these picks before finalizing curriculum changes.

Thank you